

ALAN BIJU PALAYIL

POSITION DESCRIPTION AND RESPONSIBILITIES

Current Title	Data Engineer (effective July 1, 2026)
Former Title	Application Development Analyst (December 2025 – June 2026)
Earlier Title	Associate Application Development Analyst (July 2023 – December 2025)
Employer	Genworth Financial
Department	Genworth Investments — Investment Technology
Primary Work Location	Richmond, Virginia
Travel	Occasional travel of less than 15% between Genworth's Richmond, Virginia, and Stamford, Connecticut, locations

Note: The change from Application Development Analyst to Data Engineer was a title change made as part of an organizational restructuring. Across all three titles, the core duties, systems supported, technical responsibilities, and business functions have remained substantially unchanged.

Position Summary

Develops, implements, enhances, and supports software applications, system interfaces, data pipelines, and automation solutions within Genworth Investments. The position supports the investment accounting and data-processing ecosystem used for Fixed Income, Middle Market Loans, Commercial Mortgage Loans, Derivatives, Enterprise Data Management, and related asset-accounting operations.

The role combines software development, data engineering, systems integration, and production operations. Responsibilities include requirements analysis, technical design, object-oriented programming, scripting, relational database development, data transformation, interface development, batch scheduling, data validation, reconciliation, testing, deployment, monitoring, troubleshooting, and root-cause analysis.

The position requires software-development proficiency, analytical and mathematical problem-solving skills, knowledge of relational databases and data-integration techniques, and the ability to understand investment, financial, and accounting data. The role also requires collaboration with Investments Operations, Accounting, Controllarship, business analysts, quality-assurance personnel, infrastructure teams, vendors, and onsite and offshore development teams.

Responsibilities

Software Development and System Interfaces

Develops, enhances, tests, deploys, and supports software applications used to streamline and automate processes within Genworth Investments, including Fixed Income, Middle Market Loans, Commercial Mortgage Loans, Derivatives, Enterprise Data Management, and asset-accounting operations.

Develops object-oriented and scripting-language solutions using Python, SQL, shell scripting, JSON, XML, and related technologies. Designs and maintains system interfaces, data extracts, message-generation

processes, reports, and data flows that transmit investment and accounting information between the Investments Trading and Accounting platform and ancillary financial, reporting, data-management, and vendor systems.

Develops and supports multiple production applications and interfaces. Works with project managers, business analysts, support analysts, developers, vendors, and business stakeholders to gather functional and technical requirements, evaluate system impacts, and translate detailed specifications into new or enhanced software solutions.

Applies Software Development Life Cycle practices, Agile methodologies, coding standards, version-control procedures, testing requirements, release-management processes, and enterprise change-management controls.

Data Engineering, Integration, and Automation

Designs, develops, and maintains data pipelines, ETL and data-transformation workflows, automated system interfaces, data extracts, reconciliation processes, and batch-processing solutions supporting investment accounting operations.

Provides technical support for daily, monthly, and period-end data loads, extracts, accounting feeds, interfaces, and reconciliation processes. Ensures that data is accurately and reliably transmitted to the Investments Trading and Accounting platform, enterprise data stores, data warehouses, reporting systems, vendor platforms, and downstream accounting applications.

Uses Microsoft SQL Server Management Studio to query and manage data across Markit EDM databases and Azure serverless SQL tables. Develops SQL queries, Python scripts, shell scripts, transformations, validation processes, and automated jobs used for data extraction, processing, reconciliation, and reporting.

Develops and supports cloud-based processing solutions using Microsoft Azure, Azure Synapse Analytics, Azure Data Lake Storage, notebooks, and related data-integration services. Designs and maintains JSON-based message-generation services and data-processing utilities supporting Commercial Mortgage Loan and Derivatives workflows.

Automates file transfers, job dependencies, retries, monitoring, error notifications, data validation, and operational controls. Uses enterprise scheduling tools, including AppWorx, to schedule, coordinate, monitor, and troubleshoot automated jobs and dependent workflows.

Supports secure data exchange between internal applications, cloud platforms, external vendors, and downstream systems using secure file-transfer technologies, application interfaces, and controlled batch-processing workflows.

Investment Accounting Platform Responsibilities

Supports the Investments Trading and Accounting platform, PAM (ePAM), and its related data ecosystem. Provides technical ownership and operational support for Commercial Mortgage Loan workflows within

PAM, including data interfaces, loan and collateral data, fees, payments, accounting feeds, validations, reconciliations, and downstream processing.

Supports Derivatives processing, including legacy-platform modernization, cloud migration, data transformations, message generation, accounting journals, data feeds, and reconciliation activities.

Coordinates enhancements, production support, issue resolution, testing, deployment, and ongoing maintenance for applications and interfaces supporting investment accounting operations.

Works with business and accounting stakeholders to understand investment transactions, accounting data, processing dependencies, reporting requirements, and operational controls.

Environment Management, Testing, and Deployment

Assists with the configuration, implementation, maintenance, and support of development, testing, staging, and production environments.

Develops technical and functional test plans, test cases, scripts, validation procedures, and expected results to verify the successful design, development, integration, and implementation of software applications, interfaces, reports, data pipelines, and accounting-system deliverables.

Performs unit testing, system testing, integration testing, regression testing, data validation, reconciliation testing, and production-verification activities. Supports user acceptance testing by investigating reported defects, analyzing data discrepancies, correcting software or configuration issues, and documenting test outcomes.

Assists with application-performance analysis, batch-performance monitoring, load and stress testing, query optimization, interface optimization, and processing-time improvements.

Coordinates production releases, implementation activities, change records, deployment validation, rollback planning, and post-implementation support in accordance with enterprise release-management and change-control procedures.

Production Support and Incident Resolution

Provides production support for investment applications, system interfaces, data pipelines, scheduled jobs, accounting feeds, and related back-end processes.

Investigates, diagnoses, and resolves software defects, processing failures, data discrepancies, interface errors, database issues, file-transfer failures, scheduling problems, and user-reported incidents.

Performs root-cause analysis to identify technical, data, configuration, scheduling, dependency, and process-related causes. Implements sustainable corrective actions to improve application reliability, data accuracy, auditability, maintainability, and user experience.

Monitors production jobs, processing results, exception reports, error notifications, data-quality checks, reconciliation results, and operational metrics. Coordinates issue resolution with Investments Operations, Accounting, Controllershship, infrastructure teams, vendors, business analysts, and onsite and offshore developers.

Participates in incident management, defect tracking, ticket resolution, release support, and problem-management activities.

Process Improvement and Technical Documentation

Identifies opportunities to reduce manual intervention, eliminate recurring errors, improve processing controls, strengthen data quality, and increase operational efficiency through software development and automation.

Analyzes existing workflows and recommends changes to application code, data flows, job schedules, monitoring processes, validation procedures, and operational controls.

Develops and maintains technical specifications, architecture and data-flow documentation, interface documentation, implementation plans, test plans, operational procedures, troubleshooting guides, support documentation, and knowledge base articles.

Maintains documentation for PAM (ePAM) and related supporting systems, including application dependencies, scheduled jobs, interfaces, file-transfer processes, data mappings, support procedures, and production-recovery steps.

Participates in sprint planning, backlog refinement, technical-design discussions, estimation, code review, testing, release planning, retrospectives, and continuous-improvement activities.

Technologies, Tools, and Methodologies

The position uses or supports the following technologies, tools, and methodologies:

- Python
- SQL
- Shell scripting
- JSON and XML
- Microsoft Azure
- Azure Synapse Analytics
- Azure Data Lake Storage
- Azure serverless SQL tables
- Microsoft SQL Server Management Studio
- PostgreSQL
- Markit EDM
- Informatica
- AppWorx enterprise scheduling
- Git
- Azure DevOps
- REST APIs
- Secure file-transfer technologies
- ETL and data-transformation workflows

- Data validation and reconciliation
- Batch processing and job scheduling
- Automated monitoring, retries, and notifications
- Object-oriented programming
- Relational database development
- Agile software development
- Software Development Life Cycle
- Version control
- Unit, integration, regression, and system testing
- User acceptance testing support
- Root-cause analysis
- Incident and problem management
- Release and change management
- Technical and operational documentation

Business and Technical Skills

The position requires the ability to:

- Analyze investment, accounting, operational, and technical requirements.
- Design reliable and maintainable software and data-engineering solutions.
- Understand relational data structures, data mappings, interfaces, and processing dependencies.
- Apply analytical and mathematical reasoning to data-validation, reconciliation, and troubleshooting activities.
- Communicate technical information to business, accounting, operations, and technology stakeholders.
- Manage competing development, support, and production priorities.
- Diagnose complex software, database, data-pipeline, and batch-processing issues.
- Work independently and collaboratively with onsite, offshore, vendor, and cross-functional teams.
- Prepare clear technical documentation and operational procedures.
- Support systems subject to enterprise security, audit, regulatory, and change-management controls.

Education

Bachelor of Science in Computer and Cybersecurity Engineering and Master of Science in Cybersecurity Engineering, both conferred by Illinois Institute of Technology in Chicago, Illinois, in August 2023 through a co-terminal accelerated degree program. Enrollment in the master's program began in May 2022.

Currently pursuing a Doctor of Philosophy in Information Technology at the University of the Cumberland (in progress).

Certifications and Professional Credentials

- AWS Certification Preparation Training (BackSpace Academy via Udemy) — June 2021 (training course)
- Palo Alto Networks Cybersecurity (Coursera / Palo Alto Networks) — March 2023
- Meta Front-End Developer (Coursera / Meta) — June 2023 (component of the Meta Full-Stack Engineer capstone)
- Certificate of Competence in Zero Trust (CCZT), Cloud Security Alliance — February 2025
- Meta Back-End Developer (Coursera / Meta) — January 2026 (component of the Meta Full-Stack Engineer capstone)
- Meta Full-Stack Engineer (Coursera / Meta) — January 2026 (capstone credential encompassing the Meta Front-End and Back-End Developer certifications)
- Investment Foundations Certificate (CFA Institute) — July 2026
- Vulnerability Management Detection and Response (VMDR) Certification (Qualys) — July 2026

Employment Experience

Employed by Genworth Financial beginning in July 2023 as an Associate Application Development Analyst. Promoted to Application Development Analyst in December 2025 and subsequently received a title change to Data Engineer effective July 1, 2026, as part of an organizational restructuring. Across these positions, the core duties, systems supported, technical responsibilities, and business functions have remained substantially unchanged.

Across these positions, performs software-development, data-engineering, systems-integration, production-support, and automation responsibilities for the Investments Trading and Accounting platform and its supporting data ecosystem.

Experience includes Python and SQL development, object-oriented programming, relational database development, Microsoft SQL Server Management Studio, Markit EDM databases, Azure serverless SQL tables, Azure Synapse Analytics, Azure Data Lake Storage, data transformations, shell scripting, JSON message generation, automated job development, ETL workflows, system interfaces, data extracts, accounting data feeds, reconciliation, test-plan development, system testing, production support, incident resolution, technical documentation, and root-cause analysis.